

Sonic Storyteller: Augmenting Oral Storytelling with Spatial Sound Effects

Austin Mac

University of California, Santa Barbara
Santa Barbara, California, USA
austinrmac@gmail.com

Misha Sra

University of California, Santa Barbara
Santa Barbara, California, USA
sra@cs.ucsb.edu

ABSTRACT

Traditional oral storytelling has been predominantly limited to a live audience, augmented with voice acting, sound effects, and gestures. With new technologies, the reach of oral storytelling has expanded to allow remote listeners to tune in to radio programs and more recently to listen to podcasts and audio books, asynchronously. Digital tools offer easy-to-use applications that enable anyone the ability to not only narrate a story, but also to augment it in different ways, from adding sound effects in podcasts to narrating audio books with voice effects. In this work, we present a proof-of-concept system that allows oral storytellers to augment their storytelling with spatial sound effects. We present results from a preliminary study (n=12) that found the spatial sound effects to be vivid and immersive, indicating potential for semi-automated ways of augmenting oral storytelling.

CCS CONCEPTS

• Human-centered computing → Sound-based input / output.

KEYWORDS

spatial audio, storytelling, head tracking

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1 INTRODUCTION

While convenient for the listener, creators of remote and asynchronous storytelling content cannot easily and exhaustively capture the many qualities that make in-person storytelling so unique. Focusing on an auditory perspective, in-person storytelling is inherently performative and spatialized such as in a campfire story. Campfire storytellers can use stylistic elements such as voice inflection, onomatopoeia, background noise, and stage space to enhance the storytelling experience. One element in the success of digital oral storytelling lies in its widespread reach where the careful integration of sound effects can bring the narrative to life and heighten the overall listening experience to mimic in-person storytelling.

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Besides providing more entertainment value, sound effects can “indicate the presence of the characters,” “define the feelings of the characters,” and “create associations”. Additionally, sounds can be used to help establish the scene and help the listener imagine it more vividly. In particular, spatial sound in digital storytelling can be used to indicate distance, direction, and movement of objects in the narrative [12]. Thus, sound when added on top of a storyteller’s voice, can provide a more immersive, multi-faceted, and engaging experience for the listener [3, 13]. The timing of sound effects is a critical aspect of including them into an experience. Vococentrism, the phenomenon of prioritization of the human voice over all other sounds [10], plays an important role in deciding when to play a particular sound effect. In the context of oral storytelling, when a human voice and sound effects are played at the same time, people tend to report hearing the sound effects less [12]. To add sound to radio stories, Ge et. al. [6] developed a framework alongside a semantic inference model to retrieve keywords from text appropriate for sound effects. The task is challenging to automate because it is not precisely known when and how the connection between story text and sounds happens, without substantial data labeling. Similarly, the app Novel Effect [1] adds sound effects to children’s storybooks; however, the sound effect addition is manual, and is limited only to a set of supported books.

In this work, we present Sonic Storyteller, a semi-automated system to augment digital oral storytelling (e.g., podcasts, new stories) by generating and placing spatial sound effects into a story, an otherwise manual and time consuming task. The creative process of storytelling often involves personal expression, subjective decision-making, and the exploration of unique perspectives. A level of autonomy, which can enhance artist motivation and well-being as postulated by self determination theory (SDT) [14], and agency over the output can allow creators to convey their emotions, experiences, and visions authentically while still taking advantage of advancements in AI.

2 SYSTEM DESIGN

The end-to-end process of using Sonic Storyteller to add sound effects to a story is split into two phases: 1) automated phase to insert sound effects into a story, and 2) editing phase, where the creator makes finishing touches in a more artistic fashion.

2.1 Automated Phase

In the automated phase, Sonic Storyteller generates sound effect cues, downloads appropriate sound effects, and places the sound effects in a 3D virtual scene.

2.1.1 Generating Sound Effect Cues. See Figure 1 for pipeline. First, using the GPT-3 Davinci API, we parse the input story text in order

to extract semantic information to generate sound effects. Then we use spaCy [2] and AllenNLP [5] to extract subject, verb, and direction information from the previous result. Next we map each text sound effect (subject, verb, direction) back to its location in the story text, prioritizing mapping based off of the verb then the subject. If multiple sounds can be mapped back to one location, we randomly pick one sound. Upon a match, we find the location of the match in the story text, find the next nearest comma or period, and insert the subject-verb pair into that location.

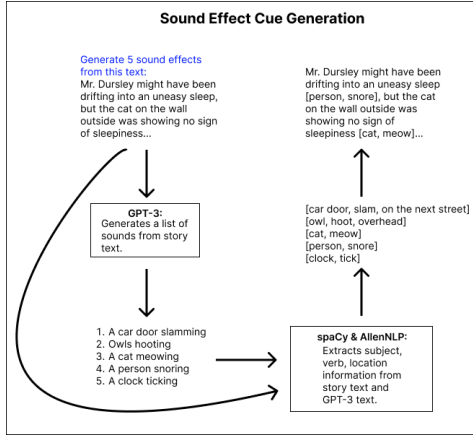


Figure 1: Sound effect cue generation pipeline. GPT-3 generates a list of text sound effects from the input story text. AllenNLP and spaCy libraries use the list of sounds and story text to create subject-verb-location tuple for each sound. Each subject-verb tuple is inserted back into the text.

2.1.2 Downloading Sound Effects. We use the sound effect cues as a guide to query the Freesound API [4] for the specified sound effect, using Freesound’s relevance score¹ as our main criteria for selecting the most appropriate sound.

2.1.3 Placing Sound Effects. Sound effects are then placed in the Unity [15] virtual three-dimensional story scene according to their associated directions in space relative to a person’s head (Figure 2).

2.2 Editing Phase

In the editing phase, storytellers can modify various sound parameters in the Unity scene including the audio source, volume, duration, quality, temporal and spatial locations through the Unity interface. Figure 2 illustrates the Unity scene in which sounds are placed. While several of these parameters could be automated with AI, the manual process enables a degree of personal expression and agency over the output.

3 PILOT EVALUATION AND RESULTS

To gauge the potential impact of spatial sound effects on the listener’s experience, we created three kinds of stories with Sonic Storyteller and had each participant (n=12) listen to exactly one.

¹https://freesound.org/docs/api/resources_apiv2.html#request-parameters-text-search-parameters

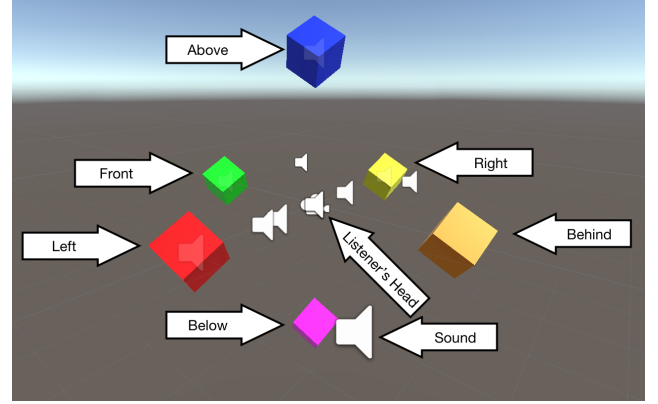


Figure 2: Unity scene for sound effect insertion. The listener’s head is located at the center of the scene. The six colored cubes are positioned relative to the listener’s head as reference points for the storytelling editor. The speaker icons represent individual sound effects.

These stories contained identical story text from *Harry Potter and the Sorcerer’s Stone*, but differed in their degree of spatial sound. The three conditions are listed below.

- Condition A: No sound effects (*No FX*)
- Condition B: Non-spatial sound effects (*Non-spatial FX*)
- Condition C: Spatial sound effects (*Spatial FX*)

Since our sample size is too small, our results are only meant to indicate the potential for further research in this area. We found that participants who listened to *Spatial FX* felt more immersed than those who listened to *Non-spatial FX*. Data was collected using four questionnaires (1: Strongly disagree - 7: Strongly agree): 1) Green & Brock Narrative Transportation Scale [7]; 2) Spatial Presence Experience Scale [9]; 3) Questions to measure cognitive load [11]; and 4) NASA Task Load Index (1: Not at all, 7: Extremely) [8]. Qualitative feedback from P9, “[i]t made the storytelling more immersive that I could feel myself with Harry.” P11 said, “I like how vivid the imagery seemed as a result of the sounds.” This indicates that spatial FX may have the potential to amplify connections to characters, imagery, and the experience of audio storytelling.

4 CONCLUSION AND FUTURE WORK

In this work, we presented Sonic Storyteller, a spatial audio application to add sound effects to oral storytelling experiences. To evaluate the output, we tested three conditions: no sound effects, non-spatial sound effects, and spatial sound effects. Results from our preliminary study show that spatial audio led to a statistically significant increase in immersion when compared to non-spatial sound effects. Our study results indicate potential for further research in adding spatial sound effects to oral storytelling. Future work includes incorporating translational motion, refining sound effect retrieval, tuning the frequency of sounds per paragraph, and performing a larger study with story creators and listeners both.

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